

Pharmacological Management of Dyslipidemia

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Session Objectives

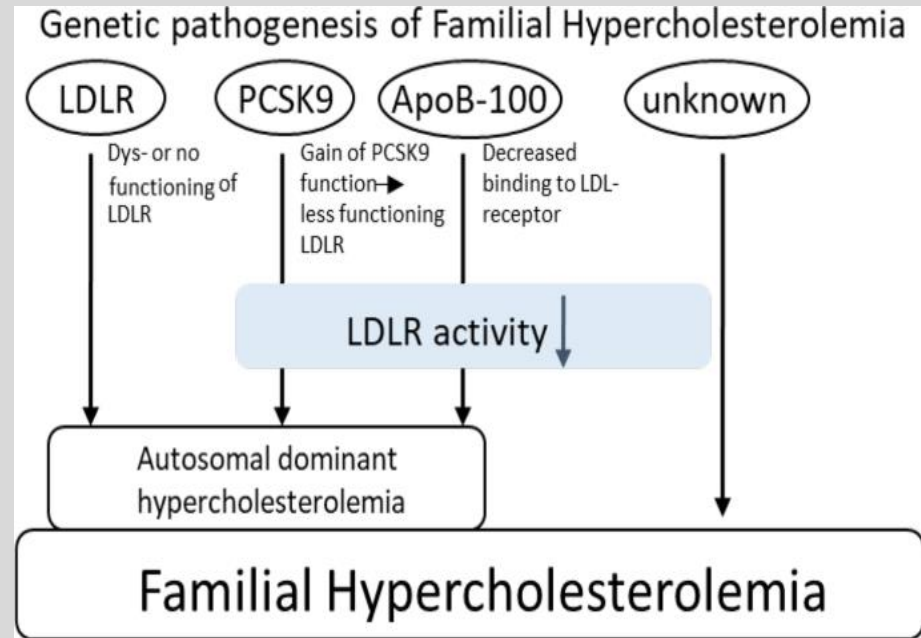
- **Define the types of dyslipidemias.**
- **Review the pathophysiology of hyperlipidemia.**
- **Describe the classes of drugs used to manage hyperlipidemia. Identify drug names and the MOA.**
- **Describe specific adverse effects and drug interactions for each antihyperlipidemic class.**
- **Identify which pharmacological mechanisms target LDL, triglycerides, or HDL levels.**
- **Understand the guidelines for the management of hyperlipidemia.**

Dyslipidemia: Summary

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- **Primary and Secondary Hypercholesterolemia**
- **Hypertriglyceridemia (> 150 mg/dL, 500 mg/dL is severe).** Increased risk of pancreatitis.
- **Familial hypercholesterolemia:**
 - **Heterozygous (LDL >190 mg/dL; pediatrics > 160 mg/dL)**
 - **Homozygous (LDL > 500 mg/dL)**



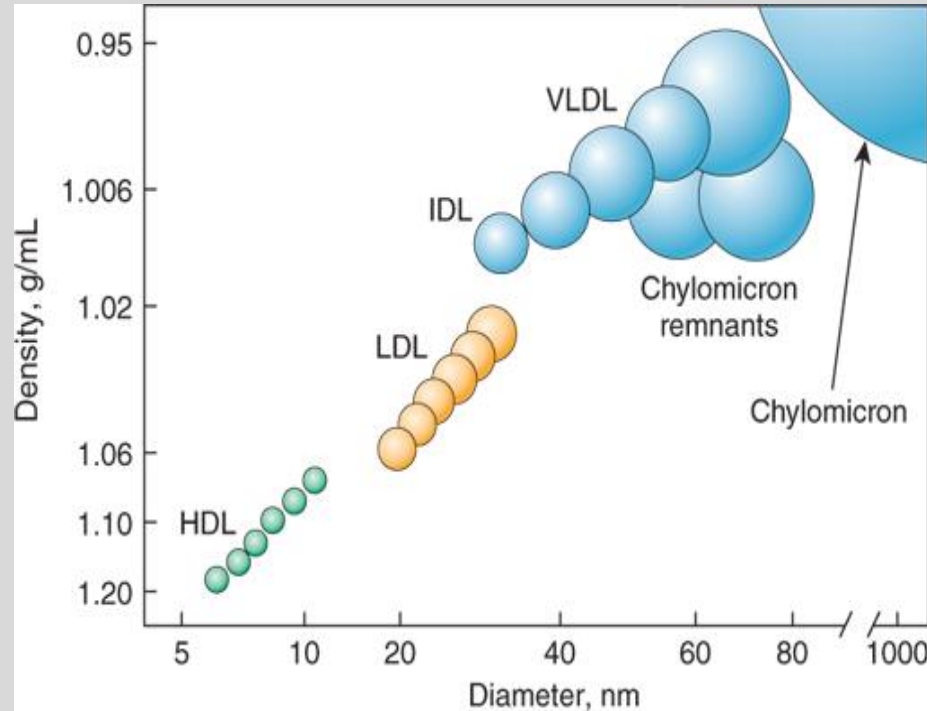
Lipoproteins

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Transport endogenous TGs and cholesterol from the liver to the tissues.

- **Chylomicrons**: Transport dietary TGs and cholesterol from the intestines to the bloodstream
- **VLDL**: very low-density lipoprotein
- **IDL**: intermediate density lipoprotein
- **LDL**: low density lipoprotein
- **HDL**: high density lipoprotein: Transport endogenous cholesterol from the tissues to the liver.



Source: Joseph Loscalzo, Anthony Fauci, Dennis Kasper, Stephen Hauser, Dan Longo, J. Larry Jameson: Harrison's Principles of Internal Medicine, 21e Copyright © McGraw Hill. All rights reserved.

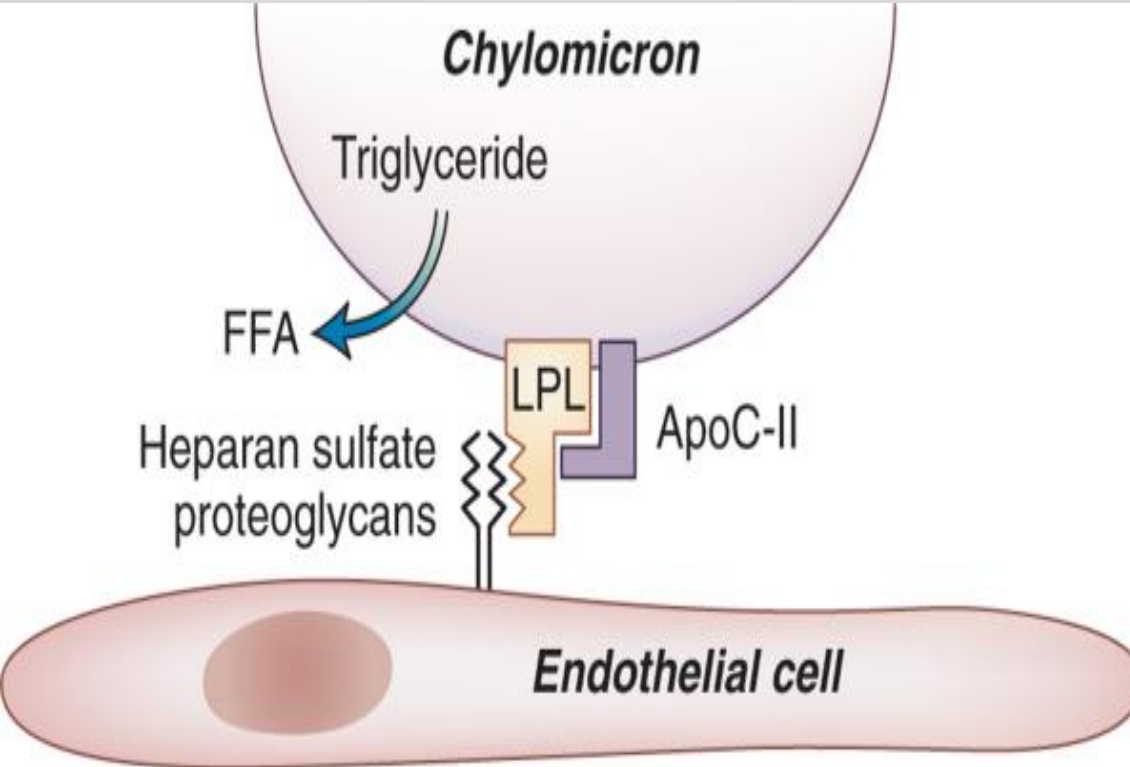
Apoproteins

Apoprotein	Lipoproteins	Function(s)
Apo B-100	VLDL, IDL, LDL	1) Secretion of VLDL from liver 2) Structural protein of VLDL, IDL, and HDL 3) Ligand for LDL receptor (LDLR) Statins?
Apo B-48	Chylomicrons, remnants	Secretion of chylomicrons from intestine
Apo E	Chylomicrons, VLDL, IDL, HDL	Ligand for binding of IDL & remnants to LDLR
Apo A-I	HDL, chylomicrons	1) Major structural protein of HDL 2) Activator of LCAT Niacin?
Apo A-II	HDL, chylomicrons	Unknown
Apo C-I	Chylomicrons, VLDL, IDL, HDL	Activation of LCAT
Apo C-II	Chylomicrons, VLDL, IDL, HDL	Activator of LPL Fibrates?
Apo C-III	Chylomicrons, VLDL, IDL, HDL	Inhibitor of LPL activity

Lipoprotein Lipase (LPL)

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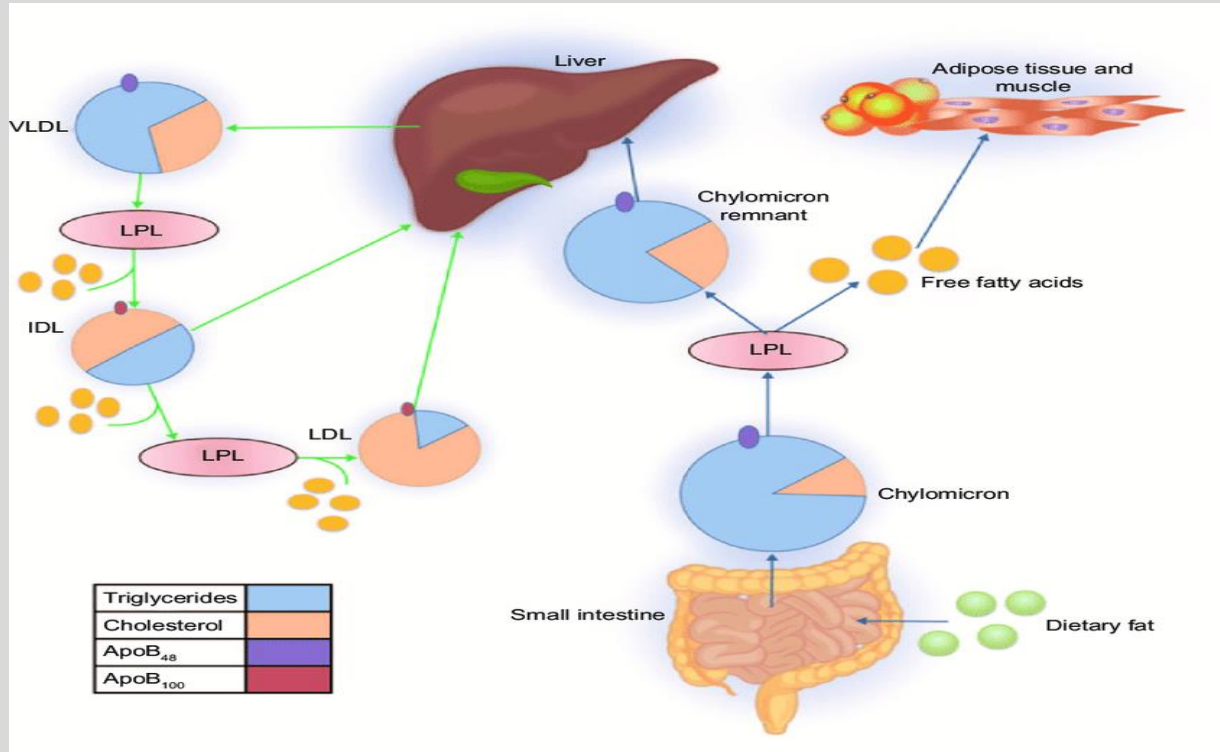


- Degrades circulating triglycerides in bloodstream
- Shrinks chylomicron
- Involved in Fibrate MOA

Lipoprotein metabolism

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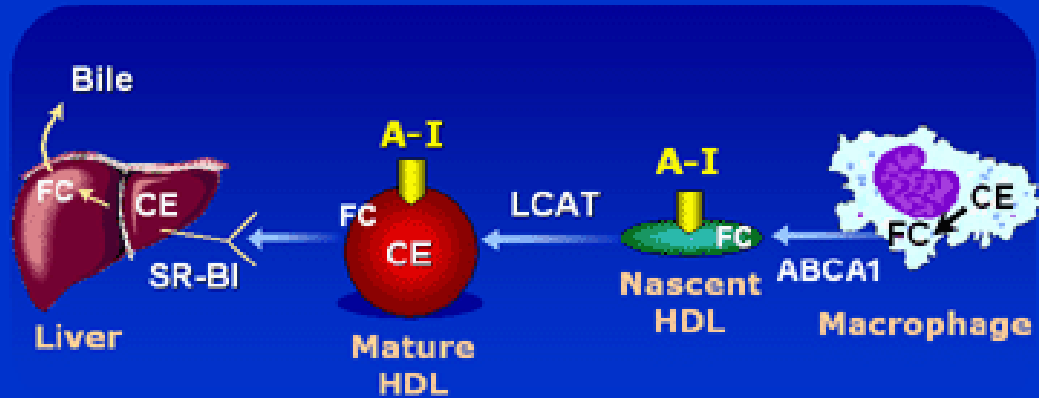
https://www.researchgate.net/figure/Overview-of-triglyceride-rich-lipoprotein-metabolism-Abbreviations-Apo-apolipoprotein_fig1_302555822

Reverse Cholesterol Transport

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HDL and Reverse Cholesterol Transport

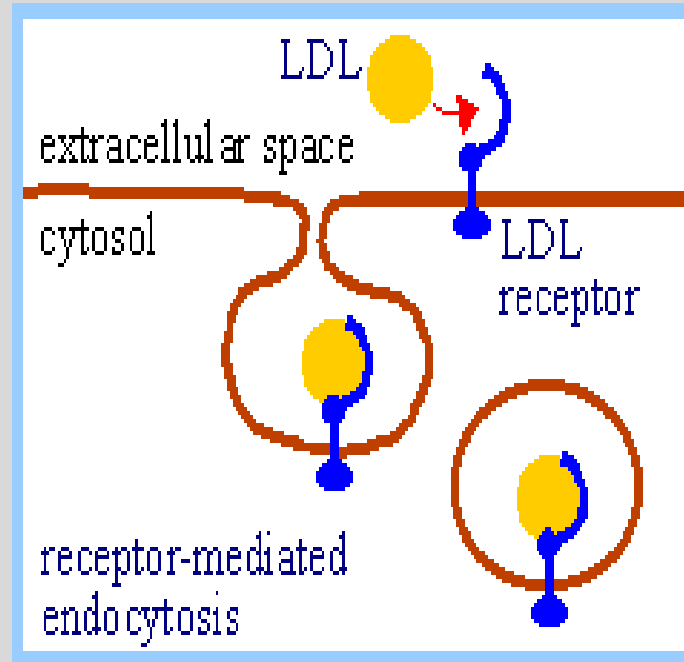
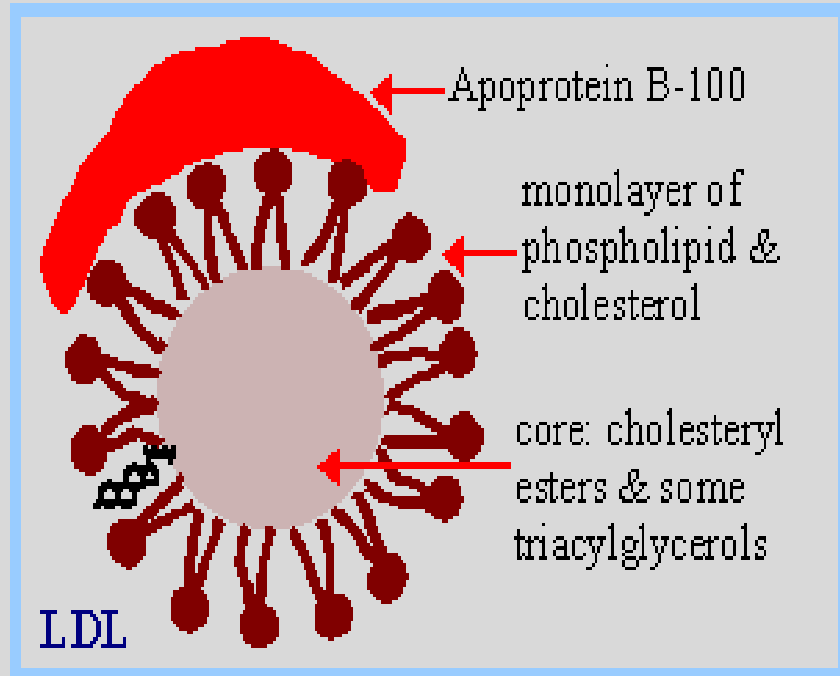


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LDL Receptor

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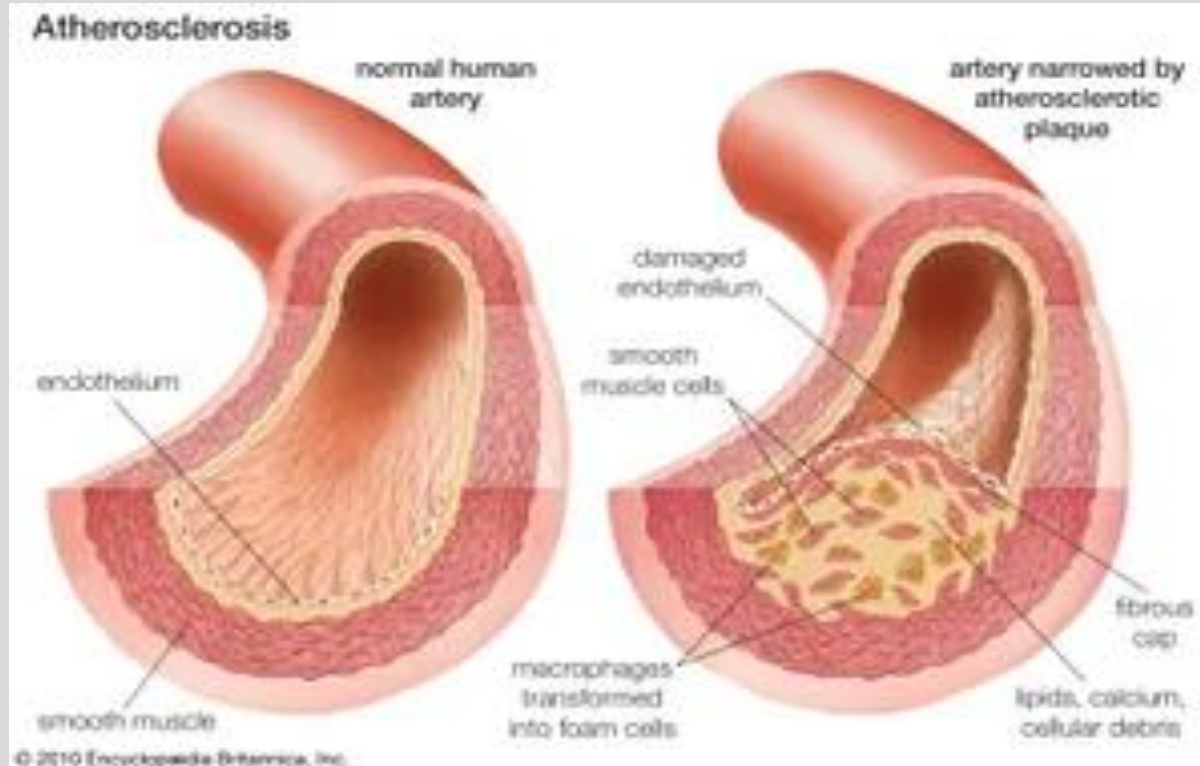
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Atherosclerosis

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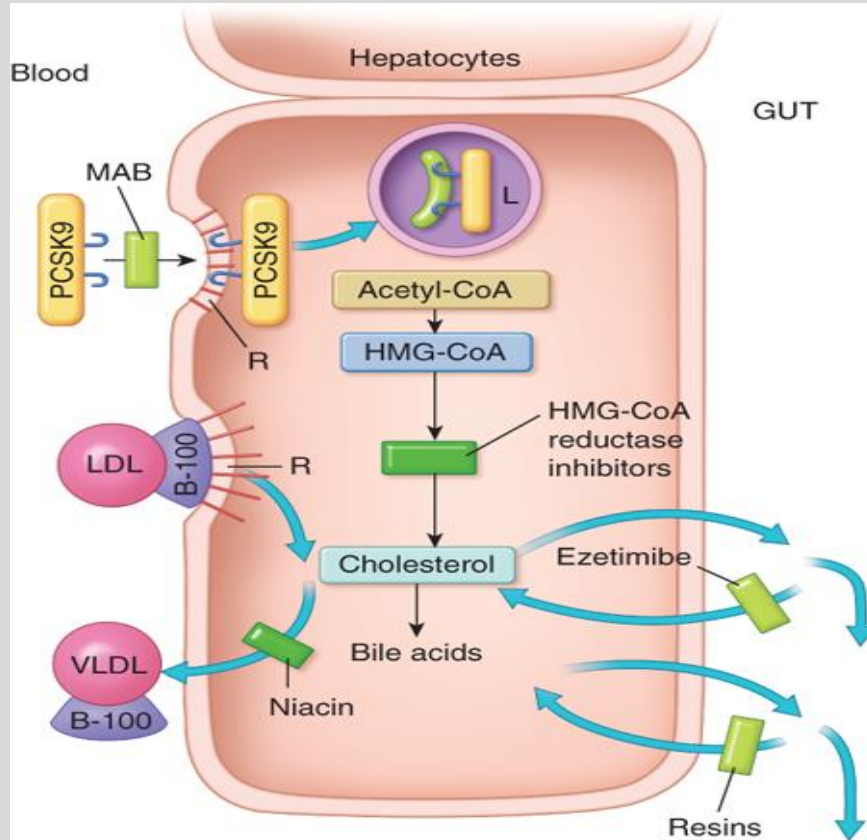
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Drugs for Hyperlipidemia

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Source: Bertram G. Katzung, Todd W. Vanderah:
Basic & Clinical Pharmacology, Fifteenth Edition
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Site of Drug Action

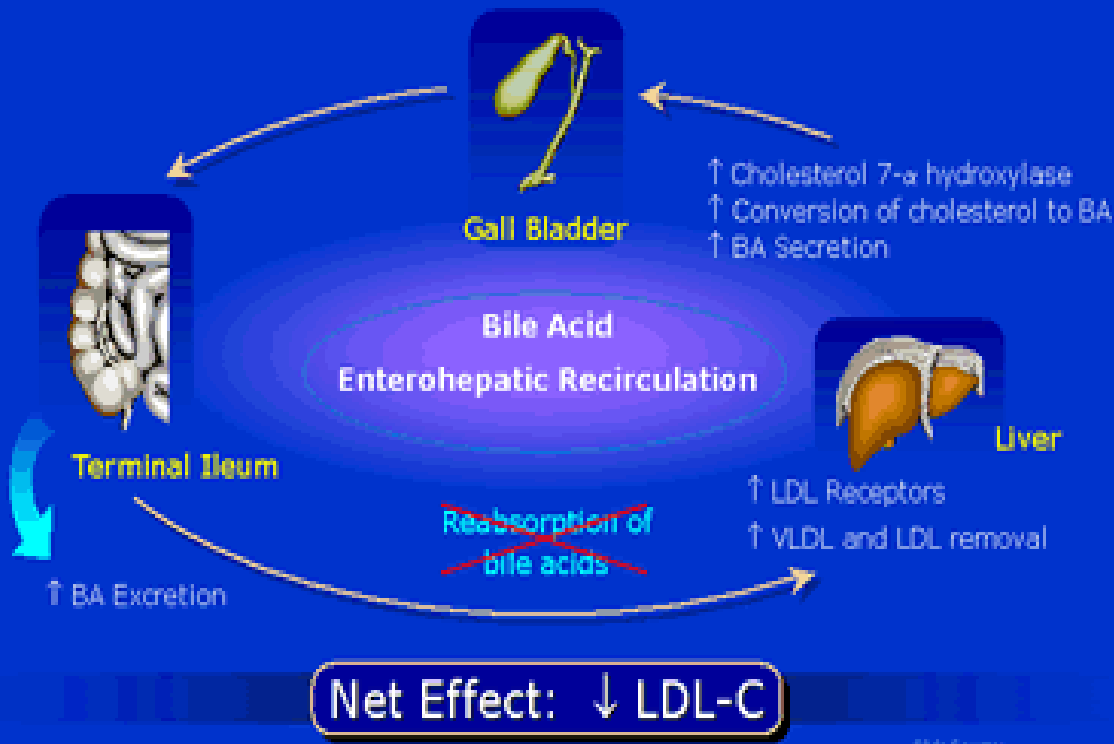
- GI: bile acid circulation (bile acid resins). Decrease cholesterol absorption (Ezetimibe).
- Liver: inhibition of cholesterol synthesis ("statins"). Inhibition of VLDL secretion (Niacin).
- Endothelial cells: activation of lipoprotein lipases (Fibrates).

Bile Acid Resins

- **Cholestyramine, Colestipol, Colesevelam**
- **Zero bioavailability: Unabsorbed**
- **MOA: prevent enterohepatic recirculation of bile acids.**
- **Increases LDL receptor formation. LDL-C blood levels decrease 30%**
- **Used in drug toxicity protocols (binds digoxin)**
- **Reduce pruritis in patients with cholestasis.**
- **Adverse effects: GI: nausea, bloating, constipation**
- **Interfere with drug absorption: thiazides, digoxin, warfarin, thyroxine, estrogens, and fat-soluble vitamins. Space from bile acid resins by several hours.**
- **Acceptable in pregnancy/peds.**

Bile Acid Resins

Bile Acid Resins: *Mechanism of Action*

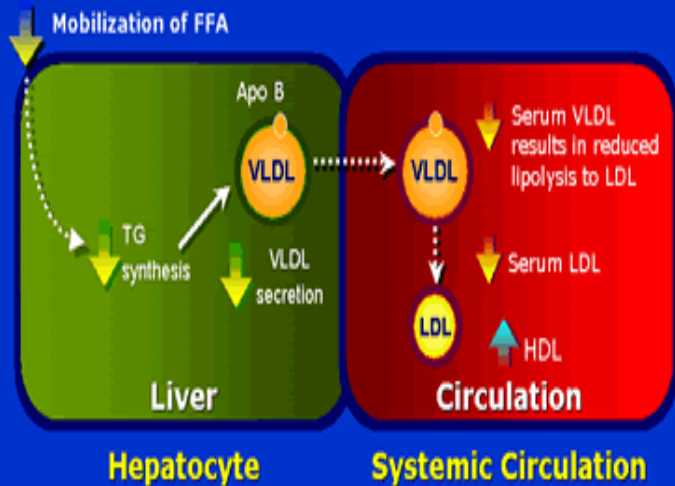


Nicotinic Acid (Niacin)

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Nicotinic Acid: Mechanism of Action



Decreases hepatic production of VLDL and of apo B

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Lipids Online Slide Library
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- **MOA:** inhibits lipolysis in adipocytes, reduces VLDL by decreasing FFA delivery to the liver. Effective against hypertriglyceridemia
- Decreases production of VLDL (35-45%), LDL (20-30%). Increases HDL (30-40%).
- **Adverse Effects:** Intense flushing and itching, reversible and preventable with aspirin, NSAID.
- Hyperglycemia, hyperuricemia, worsens PUD
- Myositis, especially if combined with statin.
- Not used in pregnancy

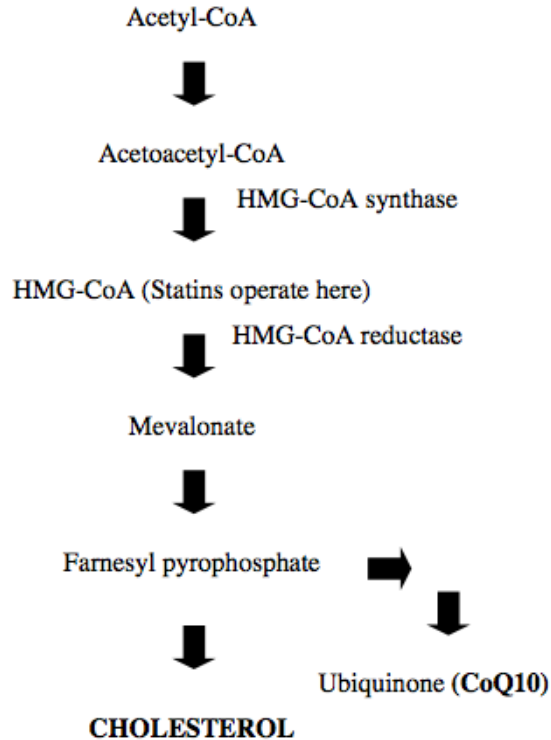
Niacin “Flush”

- **“Flushing” of the skin because there is more blood close to the surface of the body.**
- **Prostaglandin mediated vasodilation.**
- **Aspirin or NSAID prevents this effect.**
- **Slow-release formulas available (OTC: Slo-Niacin[®], Rx: Niaspan[®])**

HMG CoA Reductase Inhibitors

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- **MOA: Competitively inhibit HMG CoA reductase.**
- **Increase in the number of LDL receptors**
- **LDL decreases 25-45%.**

Scandinavian Simvastatin Survival Study (4S).

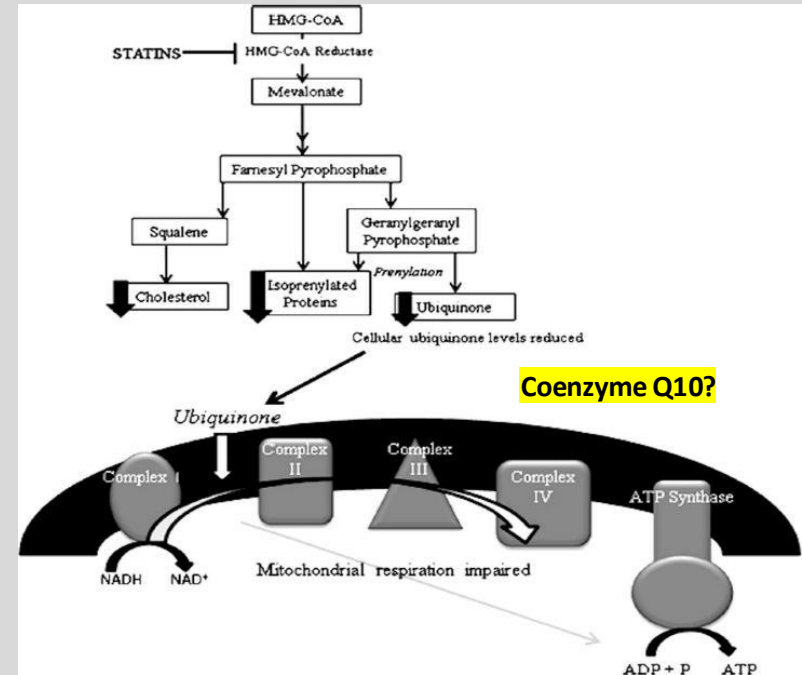
- **4444 angina/heart attack patients with cholesterol 210-310 mg/dl on simvastatin for 5.4 years.**
- **Result: LDL down 35%, deaths down 30%, cardiac deaths down 42%.**

Statin Adverse Effects

- **Hepatic toxicity** (measure aminotransferases). Caution in alcoholics, or patients with liver dysfunction.
- **Myopathy**
- **Rhabdomyolysis (creatinine kinase required)**
- Slight increase in fasting glucose levels*
- **Drug interactions**: Enzyme inhibitors (cimetidine, ketoconazole, erythromycin, grapefruit).
- Not for use in pregnancy. Pediatric patients (~ 10 years of age).
- Administer statins in the evening (except newer ones).
- **“Red yeast rice” (monacolin K)**

Myopathy

- **CYP inhibition drug interactions or combining with fibrates, niacin increases risk.**
- **Rhabdomyolysis: Breakdown of muscle fibers, releases of muscle fiber contents into the circulation. Kidney damage results.**
- **Brownish urine, elevated creatine kinase (CK).**



Statin LFT monitoring

- Aminotransferase activity should be measured at baseline, then at 8-12 weeks, but not routinely in asymptomatic patients.
- If ALT or AST concentration is consistently greater than 3 times the upper limit of normal the statin should be discontinued.
- Monitoring of liver enzymes should be more frequent if the patient is taking other drugs that have potential interactions **(CYP inhibitors)** with statins. Excessive alcohol use.

Other Considerations

- Lifestyle: high fat diet, lack of exercise, excessive alcohol intake, smoking
- Other conditions: hypothyroidism, Type II diabetes, metabolic syndrome, nephrotic syndrome
- Drugs which increase lipid levels: oral contraceptives, anabolic steroids, protease inhibitors, glucocorticoids, atypical antipsychotics, cyclosporine, retinoids.
- High homocysteine: manage by correcting Vitamin B₁₂ and folate levels.

AHA/ACC Guidelines

Current Medical Diagnosis & Treatment 2021 > Treatment of High LDL Cholesterol

Maxine A. Papadakis, Stephen J. McPhee, Michael W. Rabow

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Indications	Treatment
Presence of clinical ASCVD	High-intensity statin or moderate-intensity statin if over age 75 years
Primary elevation of LDL cholesterol ≥ 190 mg/dL (4.91 mmol/L)	High-intensity statin
Age 40–75 years Presence of diabetes LDL ≥ 70 mg/dL (1.81 mmol/L)	Moderate-intensity statin or high-intensity statin if 10-year CVD risk $\geq 7.5\%$ or other risk-enhancing criteria
Aged 40–75 years No clinical ASCVD or diabetes LDL 70–189 mg/dL (1.81–4.91 mmol/L) Estimated 10-year CVD risk $\geq 7.5\%$ or coronary artery calcium > 0 and > 75th percentile	Treat with moderate- to high-intensity statin

Statin Dosing

High-intensity Statin	Moderate-intensity Statins
Atorvastatin (40–80 mg/d) Rosuvastatin (20–40 mg/d)	Atorvastatin (10-20 mg/d) Rosuvastatin (5-10 mg/d) Simvastatin (10-20 mg/d) Pravastatin (40-80 mg/d) Lovastatin (40 mg/d)

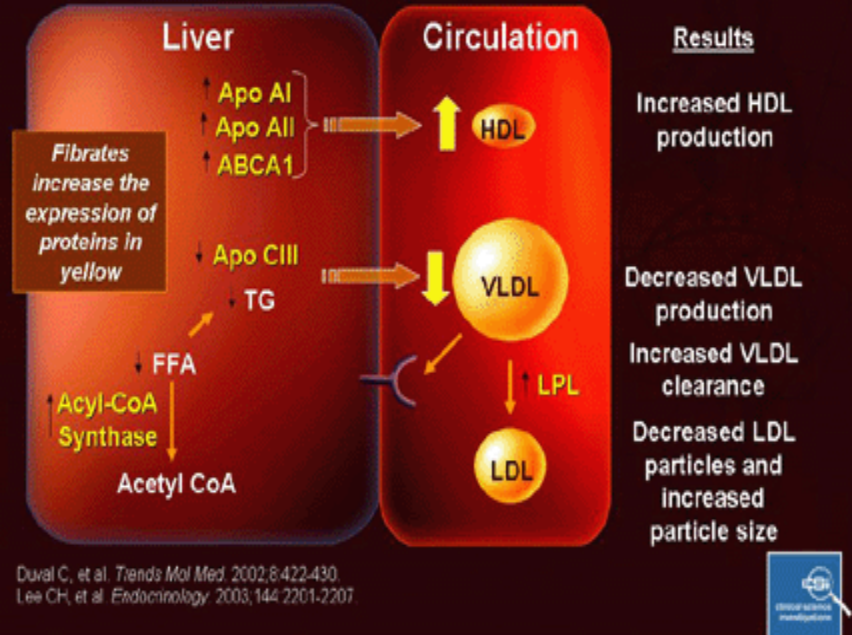
Atorvastatin and Rosuvastatin can be taken any time of day.

Fibrates

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Fibrates: Mechanisms of Action on Lipids



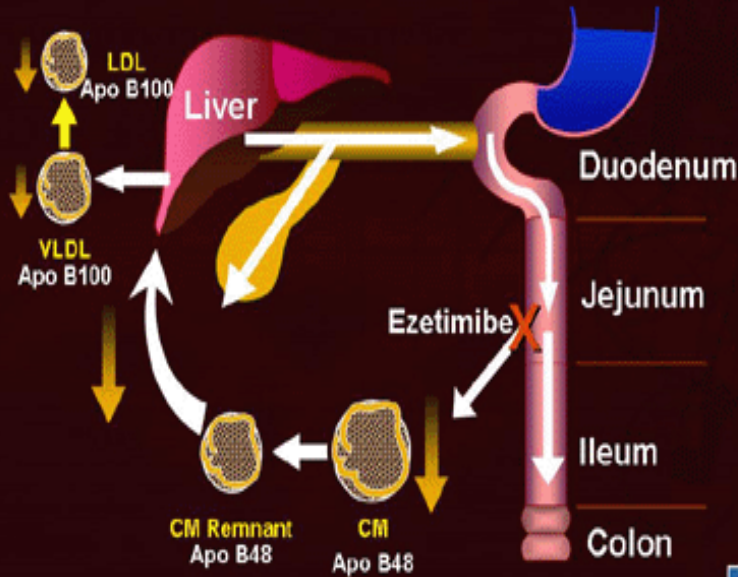
- Gemfibrozil, **Fenofibrate**.
- MOA: Ligand for PPAR α . Increases lipoprotein lipase activity and fatty acid oxidation in the liver, degrading VLDL. **Triglycerides decrease 35%.**
- Adverse effects: Gallstones
Myopathy, Rhabdo, especially with statins. Renal injury.

Ezetimibe

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Ezetimibe: Mechanism of Action



Gagne C, et al. Circulation 2002;105:2469-2475

- **MOA: Inhibits absorption of cholesterol at the brush border of the small intestine via the transporter Niemann-Pick C1-Like1 (NPC1L1).**
- **Reduction of LDL in combination with statins (watch LFTs) is 25%**
- **Adverse effects: Small increase in hepatic dysfunction, GI effects.**

PCSK9 Inhibitors

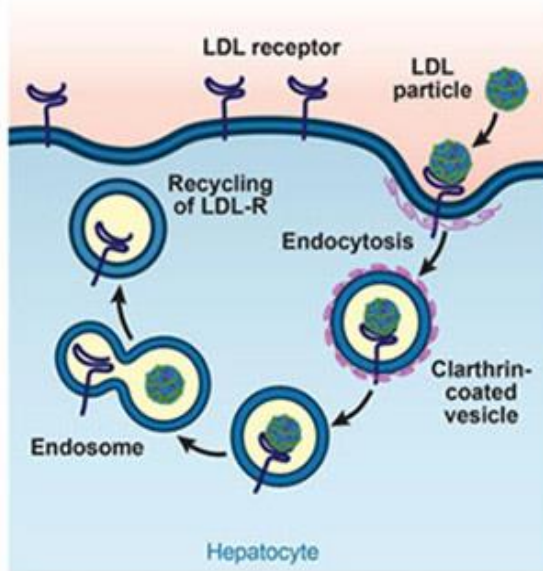
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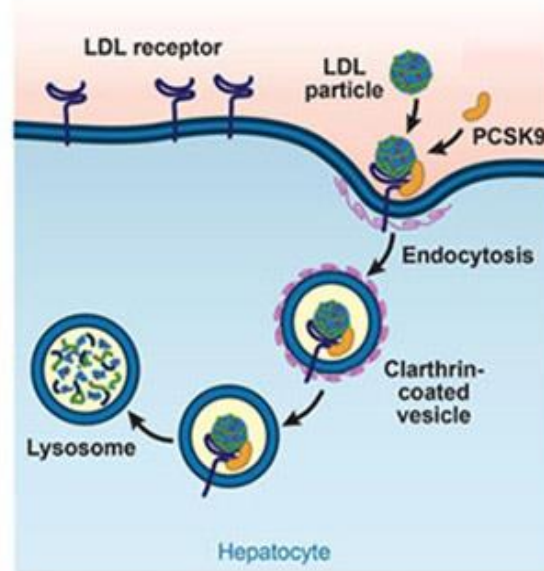
- Proprotein convertase subtilisin/kexin type 9 (PCSK9); **alirocumab, evolocumab** ” SQ dosing Q 2weeks or monthly”
- Decreased LDL receptor degradation
- For patients with familial hypercholesterolemia or clinical atherosclerotic cardiovascular disease who require additional reduction of LDL. Given with modified diet and maximal tolerated statin. LDL and APO-100 reduction
- Adverse effects: Hypersensitivity, injection site reactions, myalgia, neurocognitive effects????

PCSK9 Mechanism of Action

Drugs



LDL degradation and recycling of LDLR



PCSK9-mediated degradation of LDLR

Inclisiran

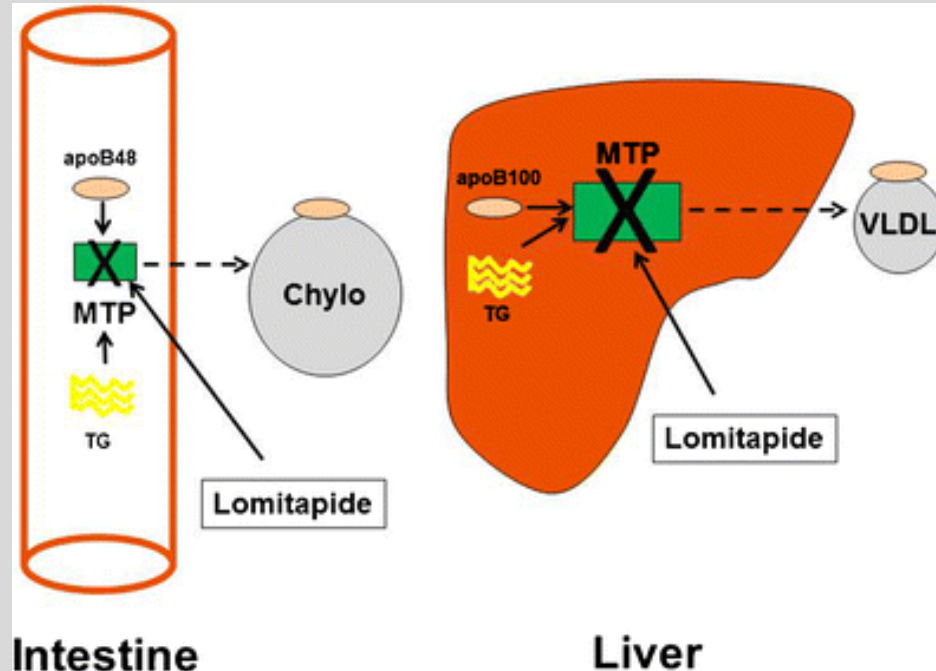
- Binds to mRNA of PCSK9 to stop gene expression.
- Result: Increased LDL receptors.
- SQ twice a year

Lomitapide

- Homozygous FH (LDL>500) not responsive to PCSK9i.
- MOA: Inhibits assembly of apoB lipoproteins in hepatocytes and intestines by inhibition of MTP.
- Adverse effects: hepatotoxicity, pregnancy X, CYP3A4 interactions. Increased toxicity if combined with statins. Restricted risk evaluation to be performed.

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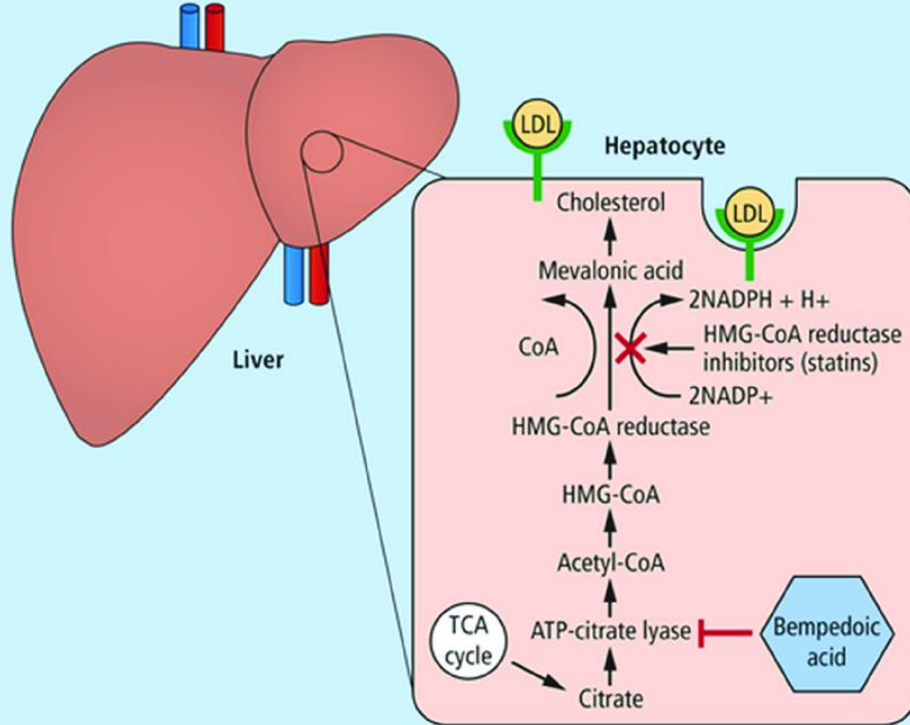
Omega-3 Fatty acids

- Stimulation of lipoprotein lipase allows degradation of VLDL (PPAR alpha agonist)
- Severe (500mg/dL) hypertriglyceridemia
- Anti-inflammatory, antithrombotic
- Adverse effect: Bleeding
- EPA/DHA: OTC and Rx formulations
- EPA only: reduces Apo CIII (Icosapent ethyl)

Bempedoic Acid--oral

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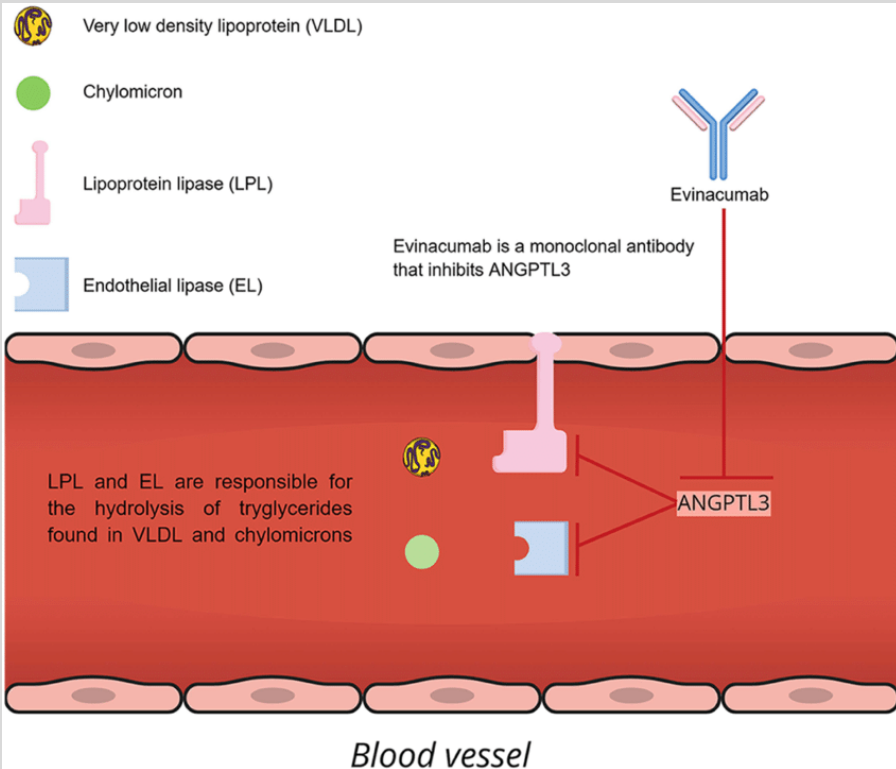


- **MOA: Inhibits ATP citrate lyase**
- **Lower risk for myopathy than statins (another option)**
- **Adverse effects: hyperuricemia, tendon rupture**

Evinacumab-monthly infusion

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- Indication: homozygous FH
- MOA: Inhibits angiopoietin-like-3 = increase LPL and endothelial lipase
- Adverse effects: Injection site reactions, anaphylaxis

Therapeutic Summary

Drugs to reduce LDL and increase receptors

- Bile acid resins lose cholesterol.
- Statins inhibit cholesterol synthesis. Also, Bempedoic acid
- PCSK9i decrease receptor degradation
- Inclisiran binds mRNA

Drugs to reduce TGs:

- Nicotinic acid decreases VLDL (TG) assembly.
- Fibrates activate lipase to degrade VLDL (TG).

Niacin raises HDL levels

Summary Slide

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- **Describe the classes of drugs used to manage hyperlipidemia. Identify drug names and the MOA.**
- **Describe specific adverse effects and drug interactions for each antihyperlipidemic class.**

Core References:

1. **Katzung & Trevor Basic and Clinical Pharmacology, 16e, 2024; Chapter 35, Agents Used in Dyslipidemia**
2. **Pharmacological Management of Dyslipidemia (Maria A. Pino, Ph.D)**
3. **Scholar Bricks: CPG**

Review Question

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A 63-year-old male presents to the emergency department with precordial chest pain. He states that this pain is often precipitated by stress or exertion and is generally relieved quickly by rest and/or nitroglycerin. There is evidence of ischemia during stress testing. Angiography demonstrates narrowing of several major heart vessels. Which of the following would be most likely to worsen the patient's angina?

- A. Acebutolol**
- B. Atenolol**
- C. Metoprolol**
- D. Nadolol**
- E. Propranolol**

Lecture Feedback Form

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<https://comresearchdata.nyit.edu/redcap/surveys/?s=HRCY448FWYXREL4R>